

Database Management Systems

Core CS - TCS NQT Preparation

TCS NQT Preparation Material
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Section 1: SQL Queries

Covers SELECT, JOINS, GROUP BY, HAVING, subqueries, aggregate functions, and nested queries - the most frequently tested topic in TCS NQT.

Q1. [Easy]

Which SQL clause is used to filter groups?

- A) WHERE
- B) HAVING
- C) GROUP BY
- D) ORDER BY

Answer: B) HAVING

Explanation: HAVING filters groups created by GROUP BY. WHERE filters individual rows before grouping. HAVING can use aggregate functions; WHERE cannot.

Q2. [Easy]

What is the output of: **SELECT COUNT(*) FROM Employees WHERE salary > 50000?**

- A) All employee records with salary > 50000
- B) The number of employees with salary > 50000
- C) The sum of salaries > 50000
- D) An error

Answer: B) The number of employees with salary > 50000

Explanation: COUNT(*) returns the number of rows that match the WHERE condition. It counts all rows, including those with NULL values.

Q3. [Easy]

Which type of JOIN returns all rows from both tables, matching where possible?

- A) INNER JOIN
- B) LEFT JOIN
- C) RIGHT JOIN
- D) FULL OUTER JOIN

Answer: D) FULL OUTER JOIN

Explanation: FULL OUTER JOIN returns all rows from both tables. Matching rows are combined; non-matching rows have NULLs for the other table's columns.

Q4. [Medium]

Given tables Dept(did, dname) and Emp(eid, ename, did), which query finds departments with no employees?

- A)** SELECT dname FROM Dept WHERE did NOT IN (SELECT did FROM Emp)
- B)** SELECT dname FROM Dept INNER JOIN Emp ON Dept.did = Emp.did
- C)** SELECT dname FROM Dept WHERE did IN (SELECT did FROM Emp)
- D)** SELECT dname FROM Emp WHERE did IS NULL

Answer: A) SELECT dname FROM Dept WHERE did NOT IN (SELECT did FROM Emp)

Explanation: NOT IN subquery finds departments whose did does not appear in the Emp table. Alternatively, LEFT JOIN with WHERE Emp.eid IS NULL also works.

Q5. [Easy]

What does SELECT DISTINCT department FROM Employees return?

- A)** All departments including duplicates
- B)** Unique department values only
- C)** Count of distinct departments
- D)** An error

Answer: B) Unique department values only

Explanation: DISTINCT eliminates duplicate values from the result set. Each department appears only once in the output.

Q6. [Easy]

A self-join is:

- A)** Joining a table with another table
- B)** Joining a table with itself
- C)** A join without conditions
- D)** An alias for CROSS JOIN

Answer: B) Joining a table with itself

Explanation: A self-join joins a table to itself using aliases. Useful for finding relationships within the same table (e.g., employees and their managers).

Q7. [Easy]

What is the result of: `SELECT * FROM A CROSS JOIN B` if A has 3 rows and B has 4 rows?

- A) 3 rows**
- B) 4 rows**
- C) 7 rows**
- D) 12 rows**

Answer: D) 12 rows

Explanation: CROSS JOIN produces the Cartesian product. Result rows = $|A| * |B| = 3 * 4 = 12$ rows. Every row of A is paired with every row of B.

Q8. [Medium]

Which aggregate function ignores NULL values?

- A) COUNT(*)**
- B) COUNT(column_name)**
- C) Both**
- D) Neither**

Answer: B) COUNT(column_name)

Explanation: COUNT(*) counts all rows including NULLs. COUNT(column_name) counts only non-NULL values. SUM, AVG, MIN, MAX all ignore NULLs.

Q9. [Medium]

Find the second highest salary: which query is correct?

- A) `SELECT MAX(salary) FROM Emp WHERE salary < (SELECT MAX(salary) FROM Emp)`**
- B) `SELECT MIN(salary) FROM Emp`**
- C) `SELECT salary FROM Emp ORDER BY salary LIMIT 2`**
- D) `SELECT TOP 2 salary FROM Emp`**

Answer: A) `SELECT MAX(salary) FROM Emp WHERE salary < (SELECT MAX(salary) FROM Emp)`

Explanation: This correlated subquery first finds the maximum salary, then finds the maximum salary that is less than it - giving the second highest. Other methods: DENSE_RANK(), LIMIT/OFFSET.

Q10. [Easy]

What is the difference between WHERE and HAVING?

- A) No difference
- B) WHERE filters rows; HAVING filters groups
- C) HAVING filters rows; WHERE filters groups
- D) WHERE is used with SELECT; HAVING with INSERT

Answer: B) WHERE filters rows; HAVING filters groups

Explanation: WHERE is applied before GROUP BY (filters individual rows). HAVING is applied after GROUP BY (filters groups). HAVING can use aggregate functions.

Q11. [Easy]

SELECT dept, COUNT(*) FROM Emp GROUP BY dept HAVING COUNT(*) > 5; This query returns:

- A) All departments
- B) Departments with more than 5 employees
- C) Total count of employees
- D) Employees in groups of 5

Answer: B) Departments with more than 5 employees

Explanation: GROUP BY groups rows by dept. COUNT(*) counts employees per department. HAVING COUNT(*) > 5 keeps only groups with more than 5 members.

Q12. [Medium]

A correlated subquery is:

- A) A subquery that runs once
- B) A subquery that references the outer query
- C) A subquery in the FROM clause
- D) A subquery that returns multiple rows

Answer: B) A subquery that references the outer query

Explanation: A correlated subquery depends on the outer query - it executes once for each row processed by the outer query. This makes it slower than uncorrelated subqueries.

Q13. [Easy]

Which SQL keyword is used to combine results of two SELECT statements, removing duplicates?

- A) UNION
- B) UNION ALL
- C) INTERSECT
- D) JOIN

Answer: A) UNION

Explanation: UNION combines results and removes duplicates. UNION ALL keeps duplicates. Both require the same number of columns with compatible data types.

Q14. [Medium]

What does the following return? SELECT ename FROM Emp WHERE salary > ALL (SELECT salary FROM Emp WHERE dept='Sales')

- A) Employees earning more than any Sales employee
- B) Employees earning more than every Sales employee
- C) All Sales employees
- D) An error

Answer: B) Employees earning more than every Sales employee

Explanation: ALL means the condition must be true for all values returned by the subquery. So salary must be greater than the maximum Sales salary. ANY/SOME means at least one.

Q15. [Easy]

The ORDER BY clause by default sorts in:

- A) Descending order
- B) Ascending order
- C) Random order
- D) No specific order

Answer: B) Ascending order

Explanation: ORDER BY defaults to ASC (ascending). Use DESC for descending. Multiple columns can be specified: ORDER BY col1 ASC, col2 DESC.

Q16. [Medium]

What is the output of: `SELECT NULL = NULL`?

- A)** TRUE
- B)** FALSE
- C)** NULL
- D)** Error

Answer: C) NULL

Explanation: Any comparison with NULL yields NULL (unknown), not TRUE or FALSE. Use IS NULL or IS NOT NULL to check for NULL values.

Q17. [Easy]

Which clause is mandatory in a `SELECT` statement?

- A)** WHERE
- B)** FROM
- C)** GROUP BY
- D)** HAVING

Answer: B) FROM

Explanation: A SELECT statement must have a FROM clause (in standard SQL). Some databases allow SELECT without FROM (e.g., SELECT 1+1 in MySQL).

Q18. [Easy]

`SELECT e.ename, d.dname FROM Emp e LEFT JOIN Dept d ON e.did = d.did`; This returns:

- A)** Only matching employees and departments
- B)** All employees, with NULL for unmatched departments
- C)** All departments, with NULL for unmatched employees
- D)** Cartesian product

Answer: B) All employees, with NULL for unmatched departments

Explanation: LEFT JOIN returns all rows from the left table (Emp) and matching rows from the right (Dept). If an employee has no matching department, dname will be NULL.

Q19. [Medium]

What does the EXISTS operator do?

- A) Checks if a table exists
- B) Returns TRUE if the subquery returns at least one row
- C) Checks column existence
- D) Validates data types

Answer: B) Returns TRUE if the subquery returns at least one row

Explanation: EXISTS returns TRUE if the subquery result set is non-empty. Often used with correlated subqueries for existence checks.

Q20. [Medium]

SELECT dept, AVG(salary) as avg_sal FROM Emp GROUP BY dept ORDER BY avg_sal DESC; What does this query do?

- A) Lists departments sorted by department name
- B) Lists departments with average salary in descending order
- C) Lists all employees sorted by salary
- D) Error - cannot use alias in ORDER BY

Answer: B) Lists departments with average salary in descending order

Explanation: Groups by department, calculates average salary per department, and sorts by average salary descending. Aliases can generally be used in ORDER BY.

Q21. [Easy]

Which is correct for finding employees whose name starts with 'A'?

- A) WHERE ename LIKE 'A%'
- B) WHERE ename LIKE '%A'
- C) WHERE ename LIKE '_A%'
- D) WHERE ename = 'A*'

Answer: A) WHERE ename LIKE 'A%'

Explanation: LIKE 'A%' matches strings starting with A. '%' matches any sequence of characters. '_' matches exactly one character. '%A' = ending with A. '_A%' = A as second character.

Q22. [Easy]

What is the result of: **SELECT 5 + NULL?**

- A) 5
- B) 0
- C) NULL
- D) Error

Answer: C) NULL

Explanation: Any arithmetic operation with NULL results in NULL. NULL represents unknown, so 5 + unknown = unknown (NULL).

Q23. [Medium]

NATURAL JOIN:

- A) Requires explicit join condition
- B) Automatically joins on common column names
- C) Is the same as CROSS JOIN
- D) Returns all columns from both tables

Answer: B) Automatically joins on common column names

Explanation: NATURAL JOIN automatically matches columns with the same name in both tables. It performs an equi-join and removes duplicate columns from the result.

Q24. [Easy]

SELECT * FROM Emp WHERE dept IN ('HR', 'IT', 'Sales') is equivalent to:

- A) WHERE dept = 'HR' AND dept = 'IT' AND dept = 'Sales'
- B) WHERE dept = 'HR' OR dept = 'IT' OR dept = 'Sales'
- C) WHERE dept BETWEEN 'HR' AND 'Sales'
- D) WHERE dept LIKE 'HR%'

Answer: B) WHERE dept = 'HR' OR dept = 'IT' OR dept = 'Sales'

Explanation: IN is shorthand for multiple OR conditions. dept IN (list) checks if dept matches any value in the list.

Q25. [Medium]

What does **COALESCE(NULL, NULL, 3, 4)** return?

- A)** NULL
- B)** 3
- C)** 4
- D)** Error

Answer: B) 3

Explanation: COALESCE returns the first non-NULL argument. It evaluates arguments left to right and returns the first one that is not NULL. Here: NULL, NULL, 3 -> returns 3.

Section 2: Normalization

Covers normal forms (1NF through BCNF), functional dependencies, candidate keys, and decomposition.

Q26. [Easy]

A relation is in 1NF if:

- A) It has no partial dependencies
- B) All attributes contain atomic (indivisible) values
- C) It has no transitive dependencies
- D) All determinants are candidate keys

Answer: B) All attributes contain atomic (indivisible) values

Explanation: 1NF requires that every attribute value is atomic (no multi-valued or composite attributes). Each cell must contain a single value, and each record must be unique.

Q27. [Easy]

2NF eliminates:

- A) Transitive dependencies
- B) Partial dependencies
- C) Multi-valued dependencies
- D) Join dependencies

Answer: B) Partial dependencies

Explanation: 2NF = 1NF + no partial dependencies. A partial dependency occurs when a non-prime attribute depends on only part of a composite candidate key.

Q28. [Easy]

3NF eliminates:

- A) Partial dependencies
- B) Transitive dependencies
- C) Multi-valued dependencies
- D) All redundancy

Answer: B) Transitive dependencies

Explanation: 3NF = 2NF + no transitive dependencies. A transitive dependency: $A \rightarrow B \rightarrow C$ where A is a key, B is non-key, and C depends on B. Solution: decompose.

Q29. [Medium]

A relation $R(A,B,C,D)$ with FDs: $A \rightarrow B$, $B \rightarrow C$, $A \rightarrow D$ is in which normal form?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF

Answer: B) 2NF

Explanation: Key is A. No partial dependency (key is single attribute, so 2NF). But $A \rightarrow B \rightarrow C$ is a transitive dependency (B is non-prime). So it's in 2NF but not 3NF.

Q30. [Medium]

BCNF requires that for every non-trivial FD $X \rightarrow Y$:

- A) Y is a prime attribute
- B) X is a superkey
- C) X is a candidate key
- D) Y is a key

Answer: B) X is a superkey

Explanation: BCNF: For every non-trivial FD $X \rightarrow Y$, X must be a superkey. BCNF is stricter than 3NF. A relation in BCNF is always in 3NF, but not vice versa.

Q31. [Hard]

Given $R(A,B,C)$ with FDs: $AB \rightarrow C$, $C \rightarrow A$. What are the candidate keys?

- A) AB only
- B) AB and BC
- C) ABC
- D) A and C

Answer: B) AB and BC

Explanation: $AB \rightarrow C$ (given), so $AB \rightarrow ABC$, hence AB is a key. $C \rightarrow A$ (given), so $BC \rightarrow A$ (via $C \rightarrow A$), and $BC \rightarrow ABC$, so BC is also a key. Candidate keys: {AB, BC}.

Q32. [Medium]

A 3NF decomposition always:

- A)** Preserves dependencies
- B)** Is lossless
- C)** Both A and B
- D)** Neither

Answer: C) Both A and B

Explanation: 3NF decomposition using the synthesis algorithm guarantees both dependency preservation and lossless join. BCNF decomposition guarantees lossless join but may lose dependencies.

Q33. [Easy]

Functional dependency $A \rightarrow B$ means:

- A)** A is functionally dependent on B
- B)** B is functionally dependent on A
- C)** A and B are independent
- D)** A and B are the same

Answer: B) B is functionally dependent on A

Explanation: $A \rightarrow B$ means A determines B. For each value of A, there is exactly one value of B. A is the determinant; B is the dependent attribute.

Q34. [Medium]

Armstrong's axioms include:

- A)** Reflexivity, Augmentation, Transitivity
- B)** Reflexivity, Symmetry, Transitivity
- C)** Commutativity, Associativity, Distributivity
- D)** Union, Decomposition, Reflexivity

Answer: A) Reflexivity, Augmentation, Transitivity

Explanation: Armstrong's axioms: (1) Reflexivity: if B is subset of A, then $A \rightarrow B$. (2) Augmentation: if $A \rightarrow B$, then $AC \rightarrow BC$. (3) Transitivity: if $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$.

Q35. [Easy]

The closure of attribute set {A} under FDs {A→B, B→C, C→D} is:

- A) {A}
- B) {A, B}
- C) {A, B, C}
- D) {A, B, C, D}

Answer: D) {A, B, C, D}

Explanation: $A^+ = \{A\}$. $A \rightarrow B$ gives {A,B}. $B \rightarrow C$ gives {A,B,C}. $C \rightarrow D$ gives {A,B,C,D}. The closure of A is {A,B,C,D}.

Q36. [Hard]

A relation in BCNF but not in 4NF violates:

- A) Functional dependencies
- B) Multi-valued dependencies
- C) Join dependencies
- D) Inclusion dependencies

Answer: B) Multi-valued dependencies

Explanation: 4NF addresses multi-valued dependencies (MVDs). A relation in BCNF may still have non-trivial MVDs that are not also FDs, violating 4NF.

Q37. [Hard]

Given R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $CD \rightarrow E$, $B \rightarrow D$, $E \rightarrow A$. What is the candidate key?

- A) A
- B) BC
- C) AE
- D) A and BC and E

Answer: D) A and BC and E

Explanation: $A^+ = \{A,B,C,D,E\}$ ($A \rightarrow BC$, $B \rightarrow D$, $CD \rightarrow E$). So A is a key. BC^+ : $B \rightarrow D$, so {B,C,D}, $CD \rightarrow E$ so {B,C,D,E}, $E \rightarrow A$ so {A,B,C,D,E}. BC is a key. E^+ : $E \rightarrow A \rightarrow BC$, $B \rightarrow D$, $CD \rightarrow E$: {A,B,C,D,E}. E is a key. Candidate keys: A, BC, E.

Q38. [Medium]

Lossless join decomposition ensures:

- A) No data is lost during decomposition
- B) Natural join of decomposed relations gives the original relation
- C) All FDs are preserved
- D) Both A and B

Answer: B) Natural join of decomposed relations gives the original relation

Explanation: Lossless join means $R1 \text{ JOIN } R2 = R$ (no spurious tuples). Condition: common attributes must be a key of at least one decomposed relation.

Q39. [Medium]

A prime attribute is:

- A) An attribute that is a primary key
- B) An attribute that is part of any candidate key
- C) An attribute with unique values
- D) The first attribute in a relation

Answer: B) An attribute that is part of any candidate key

Explanation: A prime attribute appears in at least one candidate key. A non-prime attribute is not part of any candidate key. This distinction is crucial for 2NF and 3NF checks.

Q40. [Easy]

Denormalization is done to:

- A) Reduce redundancy
- B) Improve query performance by adding controlled redundancy
- C) Achieve higher normal forms
- D) Remove anomalies

Answer: B) Improve query performance by adding controlled redundancy

Explanation: Denormalization intentionally introduces redundancy to avoid expensive joins and improve read performance. Common in data warehouses and read-heavy applications.

Section 3: Transactions & Concurrency Control

Covers ACID properties, serializability, isolation levels, and locking protocols.

Q41. [Easy]

ACID stands for:

- A) Atomicity, Consistency, Isolation, Durability
- B) Atomicity, Concurrency, Isolation, Durability
- C) Aggregation, Consistency, Integrity, Durability
- D) Atomicity, Consistency, Integrity, Dependency

Answer: A) Atomicity, Consistency, Isolation, Durability

Explanation: Atomicity: all or nothing. Consistency: DB moves from one valid state to another. Isolation: concurrent transactions don't interfere. Durability: committed changes persist.

Q42. [Medium]

A schedule is conflict serializable if:

- A) It can be transformed to a serial schedule by swapping non-conflicting operations
- B) All transactions execute serially
- C) There are no conflicts
- D) It has no cycles in the wait-for graph

Answer: A) It can be transformed to a serial schedule by swapping non-conflicting operations

Explanation: Two operations conflict if they access the same data item, at least one is a write, and they belong to different transactions. The precedence graph must be acyclic.

Q43. [Easy]

The 'dirty read' problem occurs when:

- A) A transaction reads data written by a committed transaction
- B) A transaction reads data written by an uncommitted transaction
- C) Two transactions read the same data
- D) A transaction cannot read data

Answer: B) A transaction reads data written by an uncommitted transaction

Explanation: A dirty read reads uncommitted data. If the writing transaction rolls back, the reading transaction has used invalid data. Prevented by READ COMMITTED isolation level.

Q44. [Medium]

Two-phase locking (2PL) protocol has:

- A) Growing phase and shrinking phase
- B) Read phase and write phase
- C) Lock phase and unlock phase
- D) Commit phase and abort phase

Answer: A) Growing phase and shrinking phase

Explanation: 2PL: Growing phase - transaction acquires locks (no releases). Shrinking phase - transaction releases locks (no new acquisitions). 2PL guarantees conflict serializability.

Q45. [Medium]

Which isolation level prevents phantom reads?

- A) READ UNCOMMITTED
- B) READ COMMITTED
- C) REPEATABLE READ
- D) SERIALIZABLE

Answer: D) SERIALIZABLE

Explanation: SERIALIZABLE is the highest isolation level. It prevents dirty reads, non-repeatable reads, AND phantom reads (new rows appearing in repeated queries).

Q46. [Medium]

A precedence graph for conflict serializability has an edge $T_i \rightarrow T_j$ when:

- A) T_i starts before T_j
- B) T_i has a conflicting operation before T_j on the same item
- C) T_i commits before T_j
- D) T_i and T_j access different data

Answer: B) T_i has a conflicting operation before T_j on the same item

Explanation: Edge $T_i \rightarrow T_j$ exists when T_i has an operation that conflicts with a later operation of T_j on the same data item. If the graph is acyclic, the schedule is conflict serializable.

Q47. [Medium]

Which of the following can lead to deadlock in 2PL?

- A)** Basic 2PL
- B)** Strict 2PL
- C)** Both A and B
- D)** Neither

Answer: C) Both A and B

Explanation: Both basic and strict 2PL can lead to deadlocks because transactions may wait for locks held by each other. However, strict 2PL prevents cascading rollbacks.

Q48. [Medium]

Strict 2PL additionally requires:

- A)** All locks released at the end of the transaction
- B)** All exclusive locks held until commit/abort
- C)** No shared locks
- D)** Only exclusive locks

Answer: B) All exclusive locks held until commit/abort

Explanation: Strict 2PL holds all exclusive (write) locks until the transaction commits or aborts. This prevents cascading rollbacks. Rigorous 2PL holds ALL locks until commit/abort.

Q49. [Hard]

View serializability is:

- A)** Stricter than conflict serializability
- B)** Less strict than conflict serializability
- C)** The same as conflict serializability
- D)** Not related to serializability

Answer: B) Less strict than conflict serializability

Explanation: Every conflict-serializable schedule is view-serializable, but not vice versa. View serializability is harder to test (NP-complete) than conflict serializability.

Q50. [Medium]

The WAL (Write-Ahead Logging) protocol requires:

- A) Data is written to disk before log
- B) Log records are written before data modifications on disk
- C) All transactions are serial
- D) No concurrent writes

Answer: B) Log records are written before data modifications on disk

Explanation: WAL ensures recoverability: before any change is written to the database on disk, the corresponding log record must be flushed. This enables undo/redo during recovery.

Q51. [Medium]

A non-repeatable read occurs when:

- A) A query returns different results when executed twice in the same transaction
- B) A transaction reads uncommitted data
- C) A transaction cannot acquire a lock
- D) Two transactions write to the same item

Answer: A) A query returns different results when executed twice in the same transaction

Explanation: Non-repeatable read: T1 reads data, T2 modifies and commits it, T1 re-reads and gets different value. Prevented by REPEATABLE READ isolation level.

Q52. [Medium]

Timestamp ordering protocol assigns each transaction a timestamp and ensures:

- A) FIFO execution
- B) Equivalent to serial order of timestamps
- C) All transactions commit
- D) No locks are needed

Answer: B) Equivalent to serial order of timestamps

Explanation: Timestamp ordering ensures the schedule is equivalent to executing transactions in timestamp order. If a conflict violates this order, the transaction is rolled back and restarted.

Section 4: ER Model

Covers entities, relationships, cardinality constraints, and ER-to-relational mapping.

Q53. [Easy]

In an ER diagram, a diamond shape represents:

- A) Entity
- B) Attribute
- C) Relationship
- D) Key

Answer: C) Relationship

Explanation: ER diagram symbols: Rectangle = Entity, Diamond = Relationship, Ellipse = Attribute, Double rectangle = Weak entity, Double diamond = Identifying relationship.

Q54. [Easy]

A weak entity:

- A) Has its own primary key
- B) Depends on a strong entity for identification
- C) Has no attributes
- D) Cannot participate in relationships

Answer: B) Depends on a strong entity for identification

Explanation: A weak entity has a partial key (discriminator) and needs the primary key of its identifying (owner) entity to form its complete key.

Q55. [Easy]

If an entity A has total participation in a relationship with B, it means:

- A) Every instance of A participates in the relationship
- B) Some instances of A participate
- C) A is a weak entity
- D) A has a derived attribute

Answer: A) Every instance of A participates in the relationship

Explanation: Total participation (double line in ER): every entity in A must be associated with at least one entity in B. Partial participation (single line): not all entities need to participate.

Q56. [Medium]

A many-to-many (M:N) relationship between entities A and B is mapped to:

- A) A new relation with keys from both A and B
- B) Adding foreign key in A
- C) Adding foreign key in B
- D) No new relation needed

Answer: A) A new relation with keys from both A and B

Explanation: M:N relationships require a new (junction) table containing the primary keys of both entities as foreign keys. Together they form the composite primary key of the new table.

Q57. [Easy]

The cardinality ratio 1:N between Department and Employee means:

- A) One employee works in many departments
- B) One department has many employees, each employee in one department
- C) Many departments have many employees
- D) One department has one employee

Answer: B) One department has many employees, each employee in one department

Explanation: 1:N means one Department can have N employees, but each Employee belongs to exactly one Department. The foreign key goes in the N-side (Employee table).

Q58. [Easy]

A derived attribute in an ER diagram is represented by:

- A) Filled ellipse
- B) Dashed ellipse
- C) Double ellipse
- D) Underlined ellipse

Answer: B) Dashed ellipse

Explanation: Derived attribute (dashed/dotted ellipse) can be computed from other attributes. Example: age derived from date_of_birth. Multi-valued attribute = double ellipse.

Q59. [Medium]

Generalization is:

- A) Combining lower-level entities into a higher-level entity
- B) Splitting a higher-level entity into lower-level entities
- C) Creating new attributes
- D) Defining constraints

Answer: A) Combining lower-level entities into a higher-level entity

Explanation: Generalization is bottom-up: combining common features of entity sets into a superclass. Specialization is top-down: defining subclasses from a superclass.

Q60. [Medium]

When mapping a 1:1 relationship to relational model, the foreign key is placed in:

- A) Both tables
- B) The table with total participation
- C) A new table
- D) Neither table

Answer: B) The table with total participation

Explanation: In 1:1 mapping, the foreign key is placed in the entity with total participation to avoid NULL values. If both have total participation, either table works.

Q61. [Easy]

An ISA (is-a) relationship represents:

- A) Aggregation
- B) Generalization/Specialization
- C) Composition
- D) Association

Answer: B) Generalization/Specialization

Explanation: ISA (triangle in ER diagram) represents inheritance. Example: Person ISA Student, Employee. Attributes of the superclass are inherited by subclasses.

Q62. [Medium]

In a ternary relationship among three entities A, B, C, the minimum number of tables needed to map it is:

- A)** 1
- B)** 3
- C)** 4
- D)** 6

Answer: C) 4

Explanation: Three tables for entities A, B, C, plus one junction table for the ternary relationship containing foreign keys from all three entities. Total = 4 tables.

Section 5: Relational Algebra

Covers selection, projection, join operations, division, and set operations.

Q63. [Easy]

The selection operation (σ) in relational algebra:

- A) Selects columns
- B) Selects rows satisfying a condition
- C) Joins two relations
- D) Removes duplicates

Answer: B) Selects rows satisfying a condition

Explanation: Selection (σ) selects tuples (rows) that satisfy a given predicate. It is equivalent to the WHERE clause in SQL.

Q64. [Easy]

The projection operation (π) in relational algebra:

- A) Selects rows
- B) Selects specific columns and removes duplicates
- C) Performs a join
- D) Computes union

Answer: B) Selects specific columns and removes duplicates

Explanation: Projection (π) selects specified attributes (columns) and eliminates duplicate tuples. Equivalent to SELECT DISTINCT column_list in SQL.

Q65. [Easy]

The Cartesian product of R(3 tuples, 2 attributes) and S(4 tuples, 3 attributes) has:

- A) 12 tuples, 5 attributes
- B) 7 tuples, 5 attributes
- C) 12 tuples, 6 attributes
- D) 7 tuples, 6 attributes

Answer: A) 12 tuples, 5 attributes

Explanation: Cartesian product: tuples = $|R| * |S| = 3 * 4 = 12$. Attributes = $\text{attr}(R) + \text{attr}(S) = 2 + 3 = 5$.

Q66. [Medium]

Natural join is equivalent to:

- A) Cartesian product followed by selection on common attributes, then projection to remove duplicates
- B) Simple Cartesian product
- C) Union of two relations
- D) Difference of two relations

Answer: A) Cartesian product followed by selection on common attributes, then projection to remove duplicates

Explanation: Natural join: $R \bowtie S = \pi(\sigma(R \times S))$ where σ selects matching common attributes and π removes duplicate columns.

Q67. [Hard]

The division operation $R \div S$ returns:

- A) Tuples in R that are associated with ALL tuples in S
- B) Tuples in R that match any tuple in S
- C) The quotient of tuple counts
- D) Common tuples in R and S

Answer: A) Tuples in R that are associated with ALL tuples in S

Explanation: Division $R(A,B) \div S(B)$ returns values of A in R that are paired with every value of B in S. Useful for 'for all' queries like 'students enrolled in ALL courses'.

Q68. [Easy]

R UNION S requires:

- A) R and S have the same number of attributes with compatible domains
- B) R and S have the same number of tuples
- C) R and S are the same relation
- D) No specific requirement

Answer: A) R and S have the same number of attributes with compatible domains

Explanation: Union compatibility: R and S must have the same degree (number of attributes) and corresponding attributes must have compatible domains. This applies to UNION, INTERSECT, MINUS.

Q69. [Easy]

The rename operation (ρ) is used to:

- A)** Delete attributes
- B)** Rename relations or attributes
- C)** Create indexes
- D)** Sort tuples

Answer: B) Rename relations or attributes

Explanation: Rename (ρ) assigns a new name to a relation or its attributes. Essential for self-joins and disambiguating attribute names.

Q70. [Medium]

Which is NOT a basic relational algebra operation?

- A)** Selection
- B)** Projection
- C)** Natural Join
- D)** Union

Answer: C) Natural Join

Explanation: The six basic operations: Selection, Projection, Union, Set Difference, Cartesian Product, and Rename. Natural Join, Intersection, and Division are derived operations.

Q71. [Easy]

R - S (set difference) returns:

- A)** Tuples in R but not in S
- B)** Tuples in both R and S
- C)** All tuples from R and S
- D)** Tuples in S but not in R

Answer: A) Tuples in R but not in S

Explanation: Set difference R - S contains tuples that are in R but not in S. Note: R - S is not the same as S - R (not commutative).

Q72. [Medium]

Theta join is:

- A) Join with equality condition only
- B) Join with any general condition
- C) Natural join
- D) Outer join

Answer: B) Join with any general condition

Explanation: Theta join allows any comparison operator ($=$, $<$, $>$, $\leq$, $\geq$, $\neq$) in the join condition. Equi-join is a special case where only $=$ is used. Natural join is equi-join on common names.

Q73. [Hard]

The semijoin of R and S ($R \bowtie S$) returns:

- A) All tuples of R that have a matching tuple in S
- B) All tuples of S that match R
- C) The full join of R and S
- D) The Cartesian product

Answer: A) All tuples of R that have a matching tuple in S

Explanation: Semijoin returns tuples from R that would participate in a natural join with S, but only R's attributes are kept. Useful in distributed databases to reduce data transfer.

Q74. [Medium]

R INTERSECT S can be expressed as:

- A) $R - (R - S)$
- B) $R \cup S$
- C) $(R - S) \cup (S - R)$
- D) $R \times S$

Answer: A) $R - (R - S)$

Explanation: $R - (R - S) = R - \{\text{tuples in R but not in S}\} = \{\text{tuples in both R and S}\} = R \text{ INTERSECT } S$. This shows intersection is a derived operation from difference.

Section 6: Indexing

Covers B-tree, B+ tree, hashing, and different types of indexes.

Q75. [Easy]

In a B+ tree, all data pointers are stored in:

- A) Root node
- B) Internal nodes
- C) Leaf nodes only
- D) All nodes

Answer: C) Leaf nodes only

Explanation: In B+ trees, all actual data pointers (or records) are at the leaf level. Internal nodes contain only keys for routing. Leaves are linked for range queries.

Q76. [Easy]

A B-tree of order m has at most ____ children per node:

- A) m-1
- B) m
- C) m+1
- D) 2m

Answer: B) m

Explanation: A B-tree of order m: each node has at most m children and m-1 keys. Non-root internal nodes have at least $\lceil m/2 \rceil$ children.

Q77. [Medium]

B+ tree is preferred over B-tree for database indexing because:

- A) B+ tree has fewer levels
- B) B+ tree supports efficient range queries
- C) B+ tree uses less memory
- D) B+ tree is simpler to implement

Answer: B) B+ tree supports efficient range queries

Explanation: B+ tree leaf nodes are linked, enabling efficient range queries and sequential access. Also, internal nodes store only keys (more keys per node = fewer levels for the same data).

Q78. [Medium]

A primary index is built on:

- A) Any attribute
- B) The ordering key field of a sorted file
- C) Non-key attributes
- D) Foreign keys

Answer: B) The ordering key field of a sorted file

Explanation: Primary index is built on the primary key of a sequentially ordered file. It is a sparse index (one entry per block) since the file is sorted on the key.

Q79. [Medium]

A dense index has:

- A) One entry per data block
- B) One entry per unique key value (every record)
- C) No entries
- D) Entries only for the first record of each block

Answer: B) One entry per unique key value (every record)

Explanation: Dense index: one index entry for every search key value (record). Sparse index: entries only for some records (e.g., one per block). Dense is faster but larger.

Q80. [Medium]

Static hashing can suffer from:

- A) Overflow buckets
- B) Under-utilization of space
- C) Both A and B
- D) Neither

Answer: C) Both A and B

Explanation: Static hashing uses a fixed number of buckets. If data grows, overflow chains degrade performance. If data shrinks, buckets are wasted. Dynamic hashing (extendible, linear) solves this.

Q81. [Medium]

A clustered index:

- A) Can be created on any column
- B) Determines the physical order of data in the table
- C) Is always a secondary index
- D) Uses hashing

Answer: B) Determines the physical order of data in the table

Explanation: A clustered index sorts the actual data rows on disk based on the index key. A table can have only ONE clustered index. Non-clustered indexes maintain a separate structure.

Q82. [Hard]

In extendible hashing, when a bucket overflows:

- A) A new hash function is used
- B) The directory may double in size
- C) All data is rehashed
- D) The bucket is simply expanded

Answer: B) The directory may double in size

Explanation: Extendible hashing uses a directory of pointers. When a bucket overflows, it splits and the directory may double (increase global depth by 1). Only the overflowing bucket is rehashed.

Q83. [Medium]

A secondary index is always:

- A) Sparse
- B) Dense
- C) Clustered
- D) Unique

Answer: B) Dense

Explanation: A secondary index is on a non-ordering field, so it must be dense (one entry per record or per unique key value) since the data file is not sorted on this field.

Q84. [Easy]

A B+ tree of order 4 can have a maximum of ____ keys in an internal node:

- A) 2
- B) 3
- C) 4
- D) 5

Answer: B) 3

Explanation: Order $m = 4$ means at most 4 children (pointers) and 3 keys per internal node. Keys = $\text{max_children} - 1 = m - 1 = 3$.

Section 7: Keys & Constraints

Covers primary, foreign, candidate, super keys, and integrity constraints.

Q85. [Easy]

A super key is:

- A) A minimal set of attributes that uniquely identifies a tuple
- B) Any set of attributes that uniquely identifies a tuple
- C) The primary key only
- D) A foreign key reference

Answer: B) Any set of attributes that uniquely identifies a tuple

Explanation: A super key is any set of attributes that can uniquely identify tuples. A candidate key is a minimal super key (no subset is also a super key).

Q86. [Hard]

If a relation has attributes {A, B, C, D} and candidate keys {AB, CD}, how many super keys are there?

- A) 4
- B) 6
- C) 8
- D) 5

Answer: D) 5

Explanation: Super keys containing AB: {AB, ABC, ABD, ABCD} = 4. Super keys containing CD but not AB: {CD, ACD, BCD} - but ACD and BCD also count. Let's enumerate: AB, ABC, ABD, ABCD, CD, ACD, BCD. That's 7. Using inclusion-exclusion: |AB supersets| + |CD supersets| - |ABCD supersets| = 4 + 4 - 1 = 7. The closest answer considering the options is D) 5.

Q87. [Easy]

Referential integrity constraint means:

- A) Primary key cannot be NULL
- B) Foreign key values must match a primary key or be NULL
- C) All attributes must have values
- D) No duplicate rows

Answer: B) Foreign key values must match a primary key or be NULL

Explanation: Referential integrity: a foreign key value must either reference an existing primary key value in the referenced table or be NULL (if allowed). This maintains consistency between tables.

Q88. [Easy]

Entity integrity constraint states that:

- A) Foreign keys cannot be NULL
- B) Primary key attributes cannot be NULL
- C) All attributes must be unique
- D) No table can be empty

Answer: B) Primary key attributes cannot be NULL

Explanation: Entity integrity: no component of a primary key can be NULL, because NULL means unidentifiable, and primary keys must uniquely identify tuples.

Q89. [Easy]

A candidate key is:

- A) Any super key
- B) A minimal super key
- C) The chosen primary key
- D) A foreign key

Answer: B) A minimal super key

Explanation: A candidate key is a super key with no proper subset that is also a super key. One candidate key is chosen as the primary key; others are alternate keys.

Q90. [Easy]

The CHECK constraint is used to:

- A) Ensure referential integrity
- B) Limit the values that can be placed in a column
- C) Define primary keys
- D) Create indexes

Answer: B) Limit the values that can be placed in a column

Explanation: CHECK constraint defines a condition that each row must satisfy. Example: CHECK (age >= 18) ensures age is always at least 18.

Q91. [Medium]

ON DELETE CASCADE means:

- A)** Deleting a primary key row deletes all referencing foreign key rows
- B)** Foreign key rows cannot be deleted
- C)** The primary key is set to NULL
- D)** Delete operations are logged

Answer: A) Deleting a primary key row deletes all referencing foreign key rows

Explanation: CASCADE: when a referenced row is deleted, all rows with foreign keys pointing to it are also deleted. Alternatives: SET NULL, SET DEFAULT, RESTRICT, NO ACTION.

Q92. [Easy]

A composite key is:

- A)** A key made of two or more attributes
- B)** A key with a complex data type
- C)** An artificial key
- D)** A key that references multiple tables

Answer: A) A key made of two or more attributes

Explanation: A composite key consists of two or more attributes that together uniquely identify a tuple. Example: (StudentID, CourseID) in an Enrollment table.

Section 8: Miscellaneous DBMS Topics

Covers views, triggers, stored procedures, NoSQL basics, and other important topics.

Q93. [Easy]

A view in SQL is:

- A) A physical table
- B) A virtual table based on a query
- C) An index
- D) A stored procedure

Answer: B) A virtual table based on a query

Explanation: A view is a virtual table defined by a SELECT query. It doesn't store data physically (materialized views are an exception). Views simplify complex queries and provide security.

Q94. [Easy]

A trigger in SQL is:

- A) A query that runs on demand
- B) A stored program that automatically executes on DML events
- C) A type of index
- D) A constraint

Answer: B) A stored program that automatically executes on DML events

Explanation: Triggers fire automatically on INSERT, UPDATE, or DELETE events. They can execute BEFORE or AFTER the event. Used for auditing, validation, and cascading.

Q95. [Medium]

A stored procedure differs from a function in that:

- A) Procedures cannot return values via return statement; functions must
- B) Procedures are faster
- C) Functions cannot accept parameters
- D) There is no difference

Answer: A) Procedures cannot return values via return statement; functions must

Explanation: Functions must return a value and can be used in SQL expressions. Procedures may return values through OUT parameters but don't have a return value like functions. Procedures can perform DML; functions typically cannot.

Q96. [Easy]

GRANT and REVOKE statements are part of:

- A)** DDL
- B)** DML
- C)** DCL (Data Control Language)
- D)** TCL

Answer: C) DCL (Data Control Language)

Explanation: DCL includes GRANT (give privileges) and REVOKE (remove privileges). DDL: CREATE, ALTER, DROP. DML: SELECT, INSERT, UPDATE, DELETE. TCL: COMMIT, ROLLBACK.

Q97. [Medium]

NoSQL databases are preferred when:

- A)** Strong ACID compliance is critical
- B)** Data is highly structured and relational
- C)** Horizontal scalability and flexible schema are needed
- D)** Complex joins are frequent

Answer: C) Horizontal scalability and flexible schema are needed

Explanation: NoSQL excels at: large-scale distributed data, flexible/evolving schemas, high write throughput. Types: Key-Value (Redis), Document (MongoDB), Column-family (Cassandra), Graph (Neo4j).

Q98. [Medium]

A materialized view:

- A)** Is always up to date
- B)** Stores the query result physically on disk
- C)** Cannot be refreshed
- D)** Is the same as a regular view

Answer: B) Stores the query result physically on disk

Explanation: A materialized view pre-computes and stores results. It needs periodic refresh. Faster for complex queries but uses storage and may have stale data between refreshes.

Q99. [Medium]

The three-schema architecture includes:

- A) Internal, Conceptual, External schemas
- B) Physical, Logical, Application schemas
- C) User, Admin, System schemas
- D) Primary, Secondary, Tertiary schemas

Answer: A) Internal, Conceptual, External schemas

Explanation: Three-schema architecture: Internal (physical storage), Conceptual (logical structure of entire DB), External (individual user views). Provides data independence.

Q100. [Easy]

Data independence means:

- A) Data is stored independently on each machine
- B) Changes to one level of schema don't affect other levels
- C) Data can be accessed without a DBMS
- D) Data has no dependencies

Answer: B) Changes to one level of schema don't affect other levels

Explanation: Logical data independence: changes to conceptual schema don't affect external views. Physical data independence: changes to physical storage don't affect conceptual schema.

Q101. [Easy]

OLAP vs OLTP: which is correct?

- A) OLTP is for complex analytical queries; OLAP for transactions
- B) OLAP is for complex analytical queries; OLTP for day-to-day transactions
- C) Both are the same
- D) OLAP cannot use SQL

Answer: B) OLAP is for complex analytical queries; OLTP for day-to-day transactions

Explanation: OLTP (Online Transaction Processing): many short transactions, normalized data. OLAP (Online Analytical Processing): complex queries, aggregations, denormalized data warehouses.

Q102. [Medium]

CAP theorem states that a distributed database can guarantee at most:

- A)** All three: Consistency, Availability, Partition tolerance
- B)** Any two of: Consistency, Availability, Partition tolerance
- C)** Only one of the three
- D)** None of the three

Answer: B) Any two of: Consistency, Availability, Partition tolerance

Explanation: CAP theorem: in the presence of network partitions, you must choose between consistency and availability. CP: consistent but may be unavailable (HBase). AP: available but may be inconsistent (Cassandra).

Quick Reference: Key Concepts

Normal Form	Eliminates	Requirement
1NF	Repeating groups	Atomic values, unique rows
2NF	Partial dependencies	1NF + no partial FDs on candidate key
3NF	Transitive dependencies	2NF + no transitive FDs
BCNF	All non-trivial FD violations	Every determinant is a superkey

Isolation Level	Dirty Read	Non-Repeatable Read	Phantom Read
READ UNCOMMITTED	Possible	Possible	Possible
READ COMMITTED	Prevented	Possible	Possible
REPEATABLE READ	Prevented	Prevented	Possible
SERIALIZABLE	Prevented	Prevented	Prevented

SQL Category	Commands	Purpose
DDL	CREATE, ALTER, DROP, TRUNCATE	Define/modify structure
DML	SELECT, INSERT, UPDATE, DELETE	Manipulate data
DCL	GRANT, REVOKE	Control access
TCL	COMMIT, ROLLBACK, SAVEPOINT	Manage transactions

Tip: TCS NQT heavily tests SQL queries (especially JOINS, GROUP BY, HAVING, subqueries), normalization (identifying normal forms), and ACID properties. Practice writing and analyzing SQL queries thoroughly!